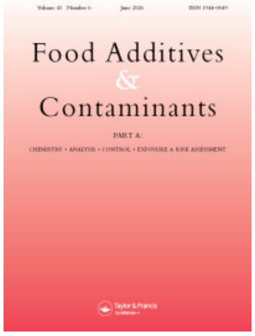

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Chromatographic fingerprinting with diode array detection for screening regulated plants with potency enhancement properties

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ABSTRACT
Analysing plant-based food supplements is particularly challenging due to their complex composition. The inclusion of multiple plant species in a single product often results in overlapping or interfering signals when conventional chromatographic methods are used. To overcome these limitations, a novel approach was applied, integrating chromatographic fingerprinting with chemometric analysis and employing a multi-wavelength detection strategy to improve resolution and accuracy. Binary classification models were built to identify the presence or absence of regulated plant materials in the samples. These models achieved strong performance and were validated through both cross-validation and external test sets, ensuring the robustness of the results. A targeted market study of 25 samples arising from illegal sources was carried out, uncovering several noteworthy and concerning findings. This study focused on supplements marketed for potency enhancement, often linked to unregulated or illicit sources. The results revealed a significant occurrence of herbal adulteration, highlighting the urgent need for systematic screening protocols and stricter regulatory measures.

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Plant food supplements; chromatographic fingerprinting; regulated plants; potency enhancement

Introduction

Erectile dysfunction (ED) can be defined as the inability to achieve and sustain an adequate penile erection for satisfactory intercourse (Wang et al. 2023). It affects a vast population of men, primarily those in the age group over 40, posing a direct impact on the mental and social well-being of the person. The aetiology of ED can be attributed to multiple factors, which include: cardiovascular disorders, diabetes, obesity, and metabolic syndrome (Leslie and Sooriyamoorthy 2024). Moreover, there is a lot of shame attached to consulting a doctor which leads to the higher acceptance of the use of potency enhancing supplements. Their popularity is considered to be a result of various advertisements through different media outlets where natural products are popularised as safe and moreover as stated above, prevents shame as such products are usually available over the counter, without a prescription. On the other hand, when considering a younger demography, these supplements have been adopted by teens and young adults as a direct result of porn consumption and media glorification, further popularising and expanding the area of concern for such products.

This paper deals with three regulated plants, *Tribulus terrestris*, *Epimedium sp*., and *Pausinystalia yohimbe*. This choice was guided by their frequent encounters in plant food supplements with potency-enhancing indications. In Belgium, the Royal decree of 1997 states that *Tribulus terrestris* and *Epimedium sp.* fall under list 3, however, the use of *Pausinystalia yohimbe* is not permitted in food supplements as it falls under list 1 of the Royal Decree ('KONINKLIJK BESLUIT van 29 AUGUSTUS 1997 Betreffende de Fabricage van En de Handel in Voedingsmiddelen Die Uit Planten of Uit Plantenbereidingen Samengesteld Zijn of Deze Bevatten (B.S. 21.XI.1997),' 1997). While list 1 deals with plants/plant parts prohibited from being sold in Belgium, list 3 consists of plants/plant parts allowed to be used, however,

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requiring a notification to the said national agencies. On the other hand, list 2 consists of edible mushrooms and provide a distinction from the poisonous varieties.

The World Health Organisation (WHO) has endorsed chromatographic fingerprinting as an official method for identifying and assessing the quality of herbs (World Health Organization 2011; Pradhan et al. 2015). Chromatographic fingerprints provide detailed chemical profiles of the samples being analysed, which can be generated using various techniques, including spectroscopic, chromatographic, or electrophoretic methods (Chen et al. 2008; Tistaert et al. 2011; Goodarzi et al. 2013). While these profiles are useful, they can sometimes lead to inaccurate identification, particularly when distinguishing between closely related species. This issue can be addressed by combining chromatographic fingerprinting with multivariate calibration methods (Kharyuk et al. 2018; Wahyuni et al. 2020; Han et al. 2022; Noviana et al. 2022; Ranjan et al. 2023; Ranjan et al. 2025). In this paper the above listed plants will be evaluated for their presence in the Belgian illegal plant food supplement market through a developed approach combining the principles of chromatographic fingerprinting and chemometrics. Their detection in complex matrices would be the primary requirement.

A chromatographic fingerprinting method was developed using reference standards for the three plants, to achieve a common analytical UHPLC–DAD method for all three plants. A multiwavelength approach was used to add maximum information to the dataset. Chemometric approaches were applied after the formation and pre-treatment of these datasets that can be imagined as data cubes with dimensions of time points × wavelengths × absorbance. Binary modelling was carried out on this dataset using the supervised technique PLS – DA. The most optimal models were validated and selected for the market study. A small market study comprising 25 samples arising from the illegal market was conducted and evaluated. This approach will help determine the current scenario concerning herbal adulteration and raise awareness among consumers.

Materials and methods

Samples and reagents

The reagents acetonitrile and methanol, both HPLC grade, along with analytical grade ethanol, were obtained from Biosolve (Valkenswaard, the Netherlands). Formic acid (FA) (99–100%) was supplied by VWR Chemicals (Leuven, Belgium). Ammonium hydroxide solution (28% w/w) was sourced from Sigma-Aldrich (Saint-Louis, Missouri, USA). Ultrapure water was generated using a Milli-Q system from Merck Millipore (Burlington, Massachusetts, USA). These solutions were used for preparing extraction solvents and mobile phase solutions for DAD analysis. Lactose was acquired from Merck (Darmstadt, Germany).

The American Herbal Pharmacopoeia (Scotts Valley, California, USA) provided the standard botanical reference materials for *Tribulus terrestris*, *Epimedium sp.*, and *Pausinystalia yohimbe*. Every reference standard came with a certificate of authenticity attesting its legitimacy. These materials were verified using botanical, macroscopic, and microscopic identification techniques in addition to thin layer chromatography.

The plant food supplements used to prepare the triturations and to conduct the limited market analysis were selected from those sent to the laboratory by the Federal agency for safety of the food chain (FASFC) and the Federal agency for medicines and health products (FAMHP) (Brussels, Belgium) to be tested for chemical adulteration. These agencies are responsible for the control and implementation of rules in respect with plant food supplements. The supplements chosen as matrix for trituration had different indications and did not claim to contain any of the three plants listed on their labels. In total, 10 plant food supplements were chosen as matrices for the triturations. They were utilised for method development and chemometric modelling. For the limited market study, 25 samples from the above-mentioned category were examined to check the presence of the target plants. All supplements used in this study tested negative for chemical adulteration.

Reference preparation

The dried material of each plant was ground with a mortar and pestle and sieved through a sieve with 70 µm pores. The ground powder was used to prepare reference solutions with a concentration of 10 mg/mL. After being weighed, the samples were put into volumetric flasks and filled with the extraction solvent. Following 30 s of vortexing and 40 min of sonication, the solutions were filtered through 0.22 µm PTFE filters and then placed into injection vials for analysis.

Preparation of triturations

Triturations were prepared in 5 varying concentrations of 1/20, 1/15, 1/10, 1/5 and 1/2. The samples for trituration preparation were selected based on the principles that they had a different indication and did not comprise or list any plant intended to be used for potency enhancement. A total of 10 + 1 (lactose) samples were selected for triturations and in total 55 triturations were prepared. The reference plant material and sample selected as blank matrices were then ground together to obtain a homogenised powder, which was stored for further use. The same method for sample preparation, as described here, was then followed for their injection into the UHPLC - DAD system.

Sample preparation

Twenty-five samples were selected for a limited market study from the samples sent in by the FASFC and FAMHP for checking for chemical adulteration. These samples were selected with following criteria: the samples are presented as potency enhancement supplements, and their claimed composition is based on a mixture of plant materials. Samples S2, S3, S6, S7, S19, and S20 claimed to contain *Tribulus terrestris* on their packaging, sample S22 had *Epimedium sp.*, and sample S24 would contain *Pausinystalia yohimbe*. These samples were prepared in the same manner as the reference solutions.

Method development and optimisation of UHPLC-DAD methodology

To enhance the extraction and detection method for reference plants, the approach was built upon the previously established HPLC-DAD technique used for identifying these regulated plants in plant food supplements (Deconinck et al. 2019). The method was adapted to UHPLC using a full factorial design, which involves evaluating all possible combinations of selected parameters to determine their effects, such as the extraction solution, the organic modifier, and the pH of the mobile phase, as described in Table 1.

The reference solutions for each analysis were prepared using seven distinct extraction solutions following the previously described sample preparation procedure. These solutions were then injected into the UHPLC-DAD system with different combinations of aqueous phases and organic modifiers. All analyses were conducted using a Waters Corporation (Milford, MA, USA) Waters Acquity UHPLC system paired with the Waters Acquity DAD. The analysis utilised an Acquity BEH Shield C18 column (2.1 mm × 100 mm, 1.7 µm).

After selection of the most effective method for extracting and detecting the reference plants, gradient optimisation was carried out to enhance peak separation and resolution. A visual assessment and counting of significant peaks determined which gradient provided the best separation and signal distribution. All subsequent experiments used this new gradient. Other parameters, such as flow rate (0.2–0.3 mL/min), injection volume (1–5 µL), and column temperature (30 °C and 35 °C), were also optimised.

Selection of wavelengths/correlation analysis

To identify the most orthogonal wavelengths that provide maximum analytical information for each reference plant, a correlation analysis was conducted using data collected through the UHPLC-DAD instrumentation.

Table 1. Parameters were tested with a full factorial design to optimise the UHPLC-DAD method, evaluating all combinations of extraction solution, organic modifier, and aqueous phase.

Extraction solutions	Aqueous phase	Organic modifier
Water	Water	Methanol
Methanol	0.1% FA and water	Acetonitrile
Methanol/Water (50:50 *v/v*)	0.1% NH_3 and water	
Acetonitrile		
Acetonitrile/Water (50:50 *v/v*)		
Ethanol		
Ethanol/Water (50:50 *v/v*)		

A correlation heatmap with wavelengths represented on both the x and y axes was created by exporting the data from Empower software and importing it into MATLAB. The dimensions of this dataset were retention time × wavelengths × absorbance. A gradient scale was employed that regarded blue as uncorrelated regions with the coefficient near 0, while yellow indicated strongly correlated regions with a coefficient close to 1. The wavelengths in the blue areas—which indicated low correlation were chosen for further analysis. Excel was then used to analyse these areas and compute correlation coefficients. A threshold of 0.97 between adjacent wavelengths was used to select less correlated wavelengths.

Data treatment

Dataset preparation

The raw data were obtained by analysing the triturations at various concentrations. A fingerprint region, which provided the most information, was selected from the reference plant, and the entire dataset was aligned to this standard dimension. The fingerprint region was found to be from 1.5 to 8 min. The selected wavelengths for each of the reference plants were then processed to obtain three datasets, each consisting of triturations followed by analysis using the set of wavelengths chosen for one of the three reference plants. These datasets were then imported into MATLAB for further chemometric treatment.

Chemometric strategy

The dataset in MATLAB consisted of a data cube of absorbance × selected wavelengths × retention time, creating a multidimensional fingerprint. Unfolding is carried out to obtain the functional dataset for chemometric analysis.

However, taking into account the differences in the data between replicates due to instrumental drift, retention time fluctuations, and other external factors, Corralized Optimised Warping (COW) was applied (Nielsen et al. 1998) as a necessary treatment before modelling. The fundamental process involves splitting both the reference and sample signals into equal segments. For each segment of the sample signal, COW permits a limited amount of stretching or compression to better align with the corresponding segment in the reference signal (Tomasi et al. 2004).

After alignment of the data using COW, datasets from each wavelength were merged into a single matrix. The DUPLEX algorithm was then applied to split the data into training (calibration) and test (validation) sets, a total of 25% of the whole data set was separated into test set while the rest was used as training set. The dataset was hence divided into 17 for test set and 50 for training set. This algorithm ensures that both subsets are representative of the overall data distribution by selecting pairs of samples exhibiting the greatest Euclidean distance, thereby maximising diversity within each set and enhancing the robustness of subsequent modelling (Chen et al. 2022). Several preprocessing techniques, such as standard normal variate (SNV) transformation, and first and second derivatives processing, were applied to possibly improve the performance of the model, along with the unprocessed data. To standardise the data, SNV adjusts the values to have a mean of zero and a standard deviation of one. This process compensates for variations in signal strength and baseline shifts, thereby improving accuracy (Barnes et al. 1989). Additionally, the first derivative enhances baseline offsets by detecting the start and end of the peak through changes in slope, whereas the second one highlights the peaks' curvature to distinguish overlapping signals and improve resolution (Guo et al. 2024).

For chemometric analysis, a supervised method - Partial Least Squares-Discriminant Analysis (PLS-DA) - was selected, since unsupervised techniques, typically used for pattern recognition, proved insufficient in handling the complexity of the dataset. PLS-DA is a useful technique especially for classifying and dealing with high dimensional datasets. In addition to the chromatographic data, the model is guided by the known class labels, which help it identify patterns that distinguish the groups (Ruiz-Perez et al. 2020).

The class labels used to perform binary modelling consisted of two categories: class 1 as positive and class 2 as negative for the targeted plant.

To determine the appropriate number of latent variables for the PLS model, a 10-fold cross-validation procedure was used on the training data. Latent variables act as features that capture the most important co-variation in the dataset with the response variable or class label. Ten-fold cross-validation helps reduce overfitting and provides a more dependable estimate of model performance. The process tested up to 30 latent variables, and the number that gave the highest (cross-) validation accuracy was selected. The final model was built using data from 55 triturated samples of each plant and an additional 11 blank matrices with 1 reference plant, making a total of 67 samples in the sample set.

The models were evaluated based on their correct classification rates through cross-validation, modelling, and external test set validation. The most optimal model was then selected to conduct the limited market study.

Software

Waters Empower 3 software (Milford, MA, USA) was employed for data acquisition from UHPLC-DAD system. Data processing and modelling were conducted using MATLAB 2019a (Mathworks, Natick, MA, USA). The ChemoAc toolbox version 4 (Brussels, Belgium) was utilised for PLS-DA application.

Results

Final chromatographic conditions

After analysing the results from the full factorial analysis, the most optimal parameters were found to be: methanol as organic modifier, 0.1% FA as aqueous phase and methanol/water (50:50) *(v/v)* as the extraction solvent. The flow rate was set at 0.25 mL/min with the column temperature adjusted to 35 °C. The injection volume was evaluated to be 2 µL. The gradient started with 90% aqueous phase (0.1% formic acid) and 10% organic modifier (methanol), for the first 50 s, followed by an increase to 30% over the next 50 s, which was held for 1.5 min. Subsequently, the organic modifier was again increased linearly to 80% over a period of 4.5 min and was held for 1 min to ensure complete elution. Finally, the gradient returned to the initial conditions of 90:10 *(v/v)* over 1 min and was held for an additional minute to re-equilibrate the column before the next run. One compromise method for all three plants was hence devised.

For validation, the repeatability of the developed method was assessed by reference plant solutions which were freshly prepared daily. These were analysed over three consecutive days, three times each day, using new vials. After gathering the chromatographic data (Figure 1), the fingerprints were visually compared with plots to assess consistency across all measurements. Chromatograms extracted from Empower software showed consistent profiles across all runs; therefore, the method was found to be repeatable. Injection repeatability was also performed and similar results were obtained.

Selected wavelengths

Colour maps, depicted in Figure 2, were created by importing the whole data cube consisting of wavelengths in the range 200–400 nm data for each reference plant into MATLAB. These colour maps were visually examined as step one and further correlation was performed in Excel. To evaluate the wavelengths that were most orthogonal, a cut-off value of 0.97 was used. Table 2 lists the wavelengths that were chosen for each standard plant material.

Chemometric analysis

PLS- DA analysis was carried out to generate binary models for each regulated plant. The 67 triturations, which consisted of 55 triturations including those of lactose, the 10 + 1 blank matrices, and the plant reference (making a total of 67 samples), were injected into the UHPLC – DAD system to generate fingerprinting data. The data set was organised to comply with the selected wavelengths and the fingerprint region (1.5-8 min) and then imported from Excel into MATLAB. The dataset was then folded to form a 3D cube consisting of 67 samples × selected wavelengths × and 7800 retention time points. As mentioned above, for *Tribulus terrestris* and *Pausinystalia yohimbe*, four characteristic wavelengths were selected, while for *Epimedium sp.*,

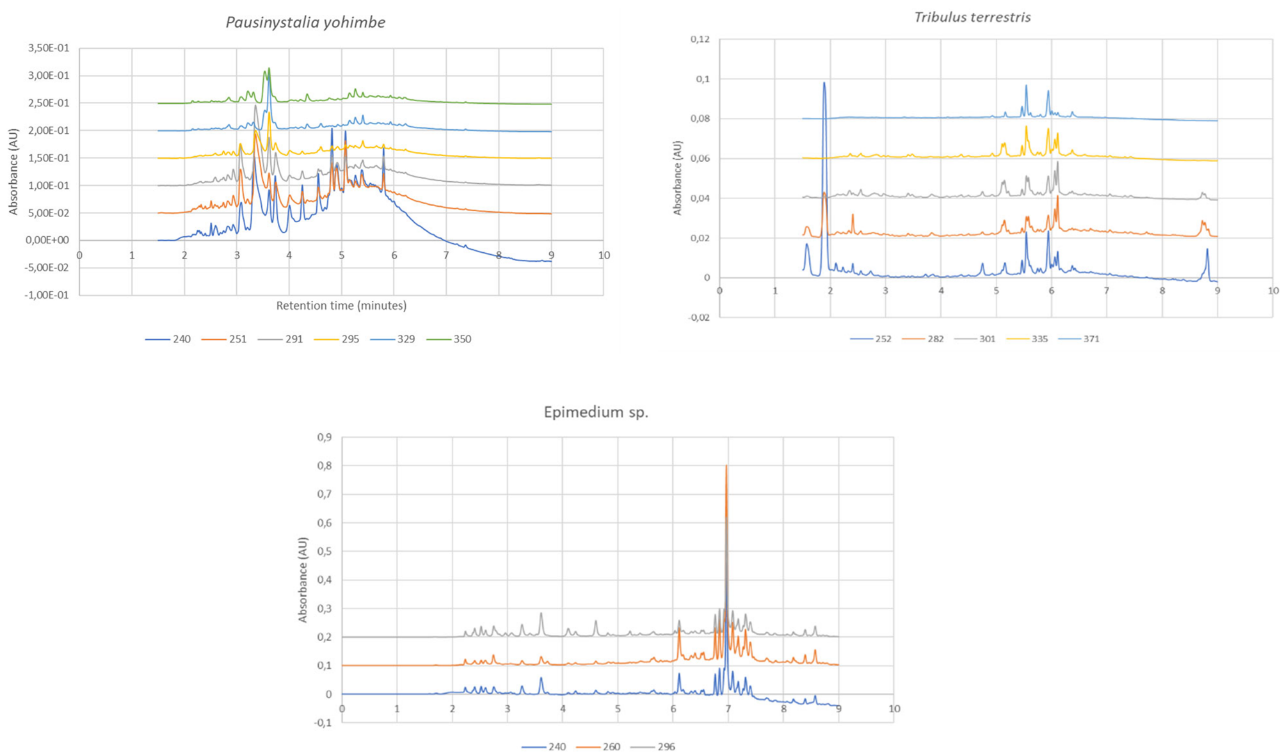

Figure 1. Overlays of the chromatograms of the concerned plants represented at their selected wavelengths (nm) with the X-axis representing retention time (in minutes) and the Y-axis representing the absorbance (in absorbance units).

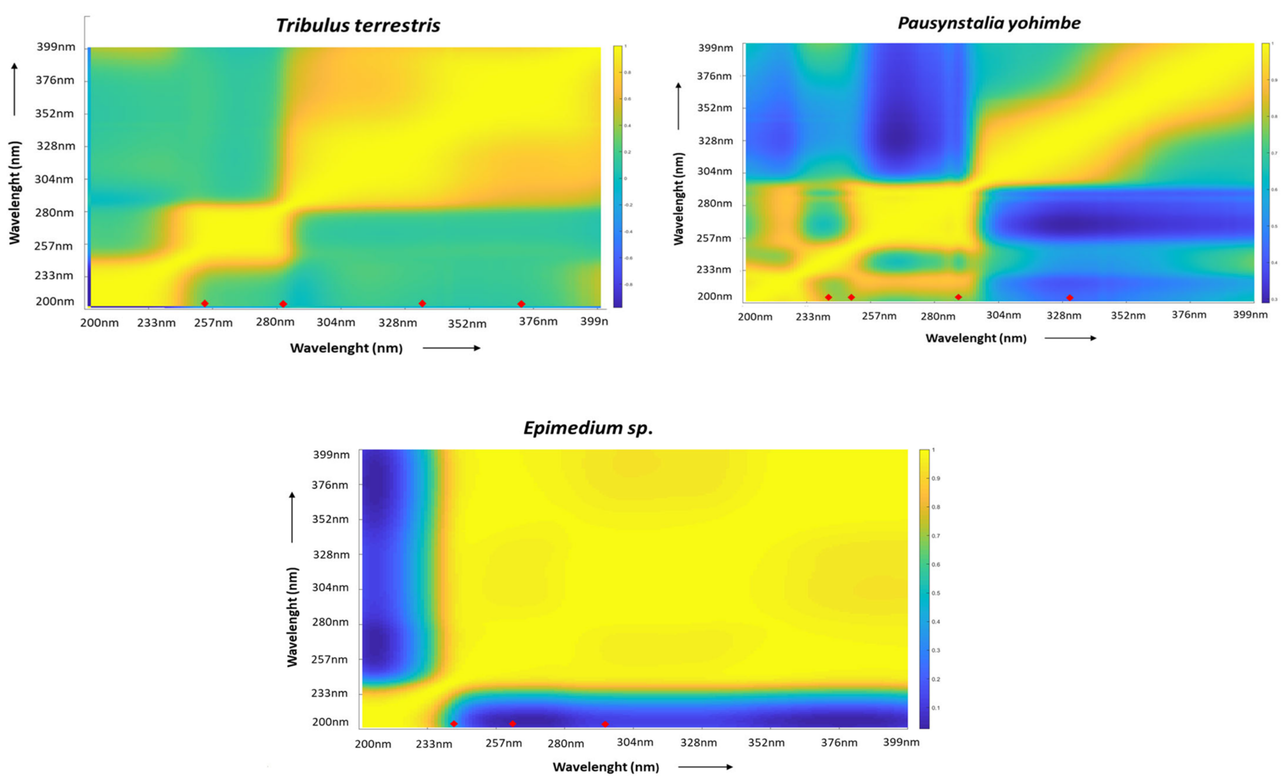

Figure 2. Colour map for *Tribulus terrestris, Pausinystalia yohimbe* and *Epimedium* sp. with wavelengths on both axis depicting the correlation between wavelengths. The gradient scale represents blue as ‘highly uncorrelated’ and yellow as ‘highly correlated’ wavelengths. The red diamond markers indicate the final selected wavelengths for each of the plant.

three wavelengths were chosen, forming this dimension to be 67×4 × 7800 for two plants and 67×3 × 7800 for *Epimedium sp.* The 3D cube was reshaped/unfolded into a 2D matrix for further treatment. As a result, the 2D data matrix dimensions were 67 samples × 31,200 variables (for *Tribulus terrestris* and *Pausinystalia yohimbe*) and 67×23,400 variables (for *Epimedium sp.*). After application of COW and preprocessing approaches, modelling was conducted using PLS-DA. The performance of the models was evaluated by the classification rates (ccr%) obtained during cross-validation, modelling, and external test set prediction. The best results acquired during the analysis of each plant data are presented in Table 3.

For *Epimedium sp.*, a ccr% of 84 was obtained for cross-validation with 19 PLS factors, indicating a correct prediction of 42 samples out of 50, where 7 were misclassified as false positives and one as a false negative. The modelling results were found to be 94% correct with two misclassifications in the external test set prediction, where one was a false positive and one a false negative.

The cross validation ccr% of *Tribulus terrestris,* after first derivative, was indicated as 82% with 16 PLS factors and a total of 41/50 correct predictions. In a similar manner to *Epimedium*, 8 were misclassified as false positives whereas 1 was depicted as false negative. The modelling results were evaluated to be 94% with only 1 misclassified sample when validated using an external test set. This sample was further assessed as a false positive.

Pausinystalia yohimbe, upon preprocessing with SNV produced a ccr% of 82 for cross validation with 13 PLS factors giving 9 misclassified samples all false positives. The modelling ccr% was evaluated as 94% and an external test set prediction also as 94% with one misclassified sample as a false positive.

It was interesting to note that the number of false positives indicates that the complexity of the matrix overpowered its ability for distinction between triturations and blank (unspiked matrices). On the other hand, the false negatives were evaluated to be those of lower concentrations (e.g. 1/20 and 1/15), which is more explainable than the previous results and indicates that the limit of detection of the approach is lying between ratios of targeted plant of 1/15 to 1/20.

A confusion matrix depicting the overall structure of the modelling results is represented in Table 4.

Table 2. Selected wavelengths for each reference plant material (nm).

Tribulus terrestris	*Pausinystalia yohimbe*	*Epimedium sp.*
252	240	240
282	251	260
335	291	296
370	329	

Market surveillance study

A limited market surveillance study comprising 25 samples was conducted with the developed strategy. These samples, as described above were received from the FASFC and FAMPH. They were seised samples, generally arising from the illegal market, that were sent to the laboratory for testing for chemical adulteration. The 25 samples selected fell into the domain of potency enhancement as their indication and were complex plant mixtures. Six of the 25 samples (Samples S2, S3, S6, S7, S19, and S20) indicated the presence of *Tribulus terrestris* on their labelling, whereas sample S24 claimed to contain *Pausinystalia yohimbe* and sample 22 depicted *Epimedium sp.* on its packaging. The developed strategy was applied to the samples and the

Table 3. Summary of PLS-DA model performance. The table includes the number of PLS factors used, ccr% for cross-validation, model training, and external test set prediction.

Potency enhancing plant	Preprocessing	PLS Factors	Cross-Validation with Misclassified Samples between brackets () (%)	Model training (ccr%)	External Test Set Validation (ccr%) with Misclassified Samples between brackets ()
Epimedium sp.	–	19	84% (8/50)	94%	88% (2/17)
Tribulus terrestris	1st derivative	16	82% (9/50)	94%	94% (1/17)
Pausinystalia yohimbe	SNV	13	82% (9/50)	94%	94% (1/17)

results regarding the presence and absence of these regulated plants are described in Table 5.

Upon application of PLS-DA, *Tribulus terrestris* was found to be present in 18 of the 25 samples. Of these, two samples, 2 and 3 that indicated the presence of *Tribulus terrestris,* were classified as negative, while samples 5–17, 19, 20, and 22–24 were predicted to be positive for the plant. This means that besides samples S6, S7, S19 and S20, claiming to contain *Tribulus terrestris,* 14 of the samples did also contain it without notification on their packaging. These results reflect the popularity of *Tribulus terrestris,* as is known in the (illegal) market (Neychev and Mitev 2016).

For *Pausinystalia yohimbe*, it was revealed that six samples (samples 2, 3, 7, 9, 10, 25) were detected as positive, whereas sample 24, which should have contained the plant, was detected as negative. This is of concern as this plant is now banned in Belgium and the EU and falls into list 3 of the Royal decree. Similarly, for *Epimedium sp.*, sample 22, which is listed on its label as *Epimedium sp.*, is predicted as negative, while six samples (samples 3, 5, 12, 20, 23, 25) labelled as negative were predicted as positive.

An interesting observation that can be made through the results is the presence of multiple plants that are detected through this strategy. This is a practice where the manufacturer, to boost the effects of the said supplements and increase their sales in the market, adds multiple potent plants. This practice clearly indicates that herbal, as chemical, adulteration is often economically motivated.

These findings further highlight the challenges in assessing illegal samples and emphasise the unreliable sources from which these samples have been obtained.

Table 4. Confusion matrix for the three regulated plants summarising the modelling for the three plants.

Potency enhancing plants	True Positives Training Set	True Positives Test Set (cv)	False Positives Training Set	False Positives Test Set (cv)	True Negatives Training Set	True Negatives Test Set (cv)	False Negatives Training Set	False Negatives Test Set (cv)
Epimedium sp.	41	15	7	1	1	1	1	1
Tribulus terrestris	40	15	8	1	0	1	1	0
Pausinystalia yohimbe	41	15	9	1	0	1	0	0

Discussion

The combination of chromatographic fingerprinting and chemometrics has been explored in this paper, and a market study for the illegal market, considering 25 samples, was performed.

Table 5. Prediction of regulated plants in illegal samples using the PLS-DA modelling

Illegal samples	Sample with claimed presence of the respective plants	Samples predicted for *Epimedium sp.,*	Samples predicted for *Tribulus terrestris*	Samples predicted for *Pausinystalia yohimbe*	Sample status as per the comparison with the label claim and modelling
1	–	–	–	–	Concordance
2	*Tribulus terrestris*	–	Absent	Present	Discrepancy
3	*Tribulus terrestris*	Present	Absent	Present	Discrepancy
4	–	–	–	–	Concordance
5	–	Present	Present	–	Finding
6	*Tribulus terrestris*	–	Present	–	Concordance
7	*Tribulus terrestris*	–	Present	Present	Discrepancy
8	–	–	Present		Finding
9	–	–	Present	Present	Discrepancy
10	–	–	Present	Present	Discrepancy
11	–	–	Present	–	Finding
12	–	Present	Present	–	Finding
13	–	–	Present	–	Finding
14	–	–	Present	–	Finding
15	–	–	Present	–	Finding
16	–	–	Present	–	Finding
17	–	–	Present	–	Finding
18	–	–	–	–	Concordance
19	*Tribulus terrestris*	–	Present	–	Concordance
20	*Tribulus terrestris*	Present	Present	–	Discrepancy
21	–	–	–	–	Concordance
22	*Epimedium sp.*	Absent	Present	–	Discrepancy
23	–	Present	Present	–	Finding
24	*Pausinystalia yohimbe*	–	Present	Absent	Discrepancy
25	–	Present	–	Present	Finding

Plant food supplements are usually composed of complex plant mixtures that make the identification of regulated plants not so straightforward. Method development was carried out using UHPLC – DAD, where a multiwavelength approach (selecting orthogonal wavelengths) was employed to include as much information as possible generated from fingerprinting in the dataset. A compromise method for all three regulated plants was devised, making it easier for future applications.

For chemometric approaches, binary modelling was employed, which constituted the classification of the regulated plant as class 1 (assigned positive for the regulated plant) and class 2 (assigned negative for the regulated plant). A satisfactory result for all three plants was obtained with one misclassification each for *Tribulus terrestris* and *Pausinystalia yohimbe* and two samples misclassified for *Epimedium sp.* when the models were validated using an external test set. Therefore, a good predictive ability for the models could be concluded.

To apply the generated models to real-life samples, a market study consisting of 25 samples arising from the illegal market was conducted. The samples were evaluated to be highly adulterated with the presence of the plant *Tribulus terrestris,* with 18 samples out of 25 being deemed as positives by the classification technique. It raises concern, but it is also representative of its market dominance over the past few years. *Epimedium* spp., on the other hand, was found to be positive in 6 samples, which also pushes it more towards the herbal adulteration narrative. On the other hand, *Pausinystalia yohimbe* was also found to be positive in 6 samples. This also raises eyebrows as it is a banned product in most European countries due to its proven toxicity (Poppenga 2006). According to the Royal Decree, it also falls under List 1, which contains prohibited plants/plant parts for use; thus, its sale is forbidden in Belgium and the EU.

For all three plants, a conclusive statement about the instances of fraud and herbal adulteration could be made. *Tribulus terrestris* was detected in 4 out of 6 samples that claimed its presence; therefore, samples 2 and 3 show cases of fraud where the ingredient list does not match the results (which indicate its absence). Furthermore, for *Epimedium sp.* and *Pausinystalia yohimbe,* it was also revealed that the claimed products did not match the label claims, again indicating the prevalence of such fraudulent cases. Whereas, for all the other samples where the regulated plants were detected to be positive without any presence on the labelling, instances of adulteration can be anticipated. This is surprising but could be expected as these illegal markets are not regulated. The possibility that a lower concentration than claimed on the labelling, or a cheaper, or morphologically similar plant could be added to this supplement remained undetected by the models.

Moreover, when the modelling results were compared, it was revealed that a few samples tested positive for multiple plants, showing a different underlying story that could involve an effort to increase and boost sales of these potency-enhancing supplements.

Even though the combination of chemometrics with chromatographic fingerprinting seems feasible, like any other technique, the results produced should always be validated through a different method, whether when the sample is from the legal market or if the analysis results are to be used in a judicial procedure. The use of next-generation sequencing or DNA metabarcoding is on the rise for the detection of plant or plant species in complex matrices (Raclariu et al. 2018; Delgado-Tejedor et al. 2021). This also comes with the realisation that not every model can be 100% accurate and should always be validated/confirmed by different approaches.

Conclusions

This methodology depicts the feasibility of chromatographic fingerprinting with chemometrics for the detection of regulated plants in potency-enhancing plant food supplements. It can be regarded as a pre-screening approach. The market study provided an overview of the real-life scenario concerning the illegal market. The results obtained indicate the importance of a screening approach that could apply to complex matrices. This approach was a step in that direction, but, as mentioned before, it should be complemented with another confirmatory technique in case of decision-making on withdrawal from the market (legal market) or a judicial context.

Conversely, the presence of adulterated products in the market was verified, demanding stricter regulation and oversight of plant food supplements and their entry channels.

Author contributions

CRediT: **Surbhi Ranjan**: Data curation, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing; **Lika Bakhturidze**: Formal analysis, Methodology, Validation; **Erwin Adams**: Conceptualization, Project administration, Supervision, Writing – review & editing; **Eric Deconinck**: Conceptualization, Funding acquisition, Project administration, Supervision, Writing – review & editing.

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